

EECS 16A Designing Information Devices and Systems I

Discussion 8B

1. Resist the Touch

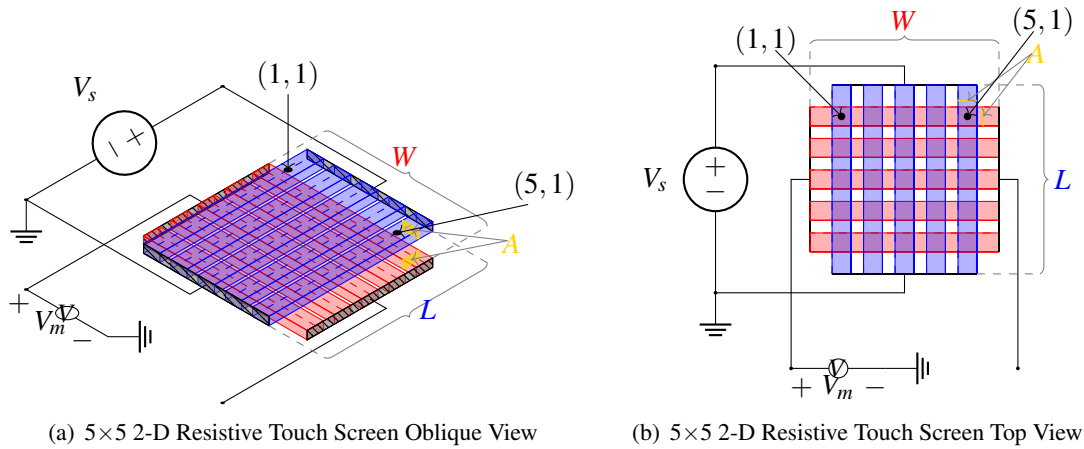


Figure 1: $N \times N$ Resistive Touch Screen, $N = 5$

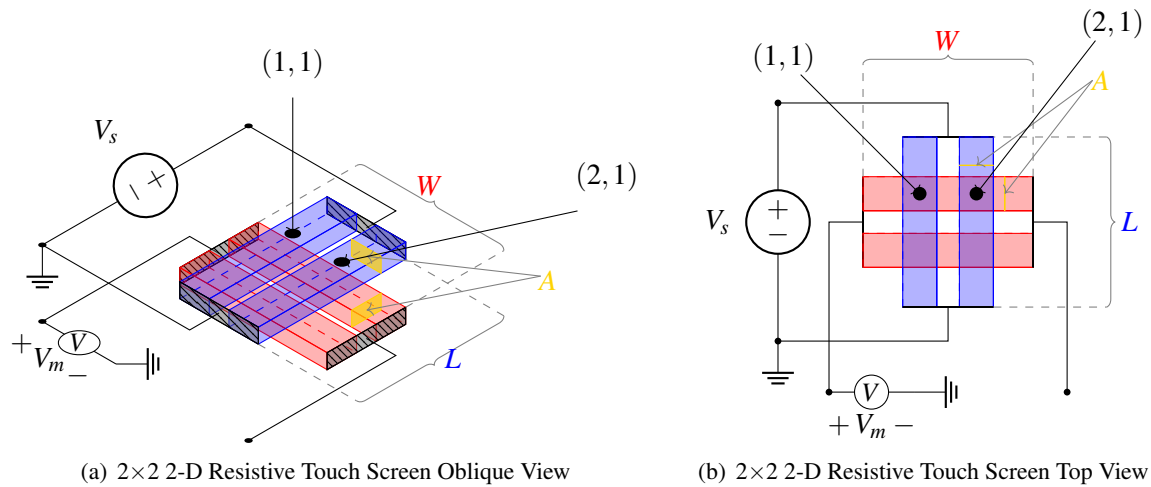
In this question we will be re-examining the 2-dimensional resistive touchscreen. This touchscreen, is slightly different to the one shown in lecture and more like the one we will be examining in lab.

The touchscreen has length L and width W and is composed of a rigid bottom-layer and a flexible top-layer. Instead of having two continuous resistive sheets on the top and bottom layers, this is a simpler implementation with N vertical strips of conductive material in the top layer and N horizontal strips of conductive material in the bottom layer. The strips of a single layer are all connected by an ideal conducting plate on each side. All strips have resistivity, ρ , and cross-sectional area, A .

Assume that all top layer resistive strips and bottom layer resistive strips are spaced apart equally, and that the upper left touch point in Figure 1(b) is position $(1, 1)$, and the upper right touch point is $(N, 1)$. The spacing (measured from the center of one strip to the center of the next) between the strips in the top layer is $\frac{W}{N+1}$, and the spacing between the strips in the bottom layer is $\frac{L}{N+1}$.

- (a) Find the resistance R_y for a single vertical blue strip and R_x for a single horizontal red strip, as a function of the screen dimensions W and L , the strip resistivity ρ , and the cross-sectional area A .

- (b) Consider a 2×2 example for the touchscreen circuit, shown in Figure 2.

Figure 2: 2×2 Resistive Touch Screen

Assume that we connect a voltage source V_s , between the top and bottom terminals of the blue strips, and a voltmeter V_m to one of the left or right terminals as depicted in the diagram.

If $V_s = 3\text{ V}$, $R_x = 2000\Omega$, and $R_y = 2000\Omega$, draw the equivalent circuit for when the point $(2,2)$ is pressed and solve for the measured voltage, V_m , with respect to ground.

Remember that all top layer resistive strips and bottom layer resistive strips are spaced apart equally, and that the upper left touch point is position $(1,1)$. The spacing between the strips in the top layer is $\frac{W}{N+1}$, and the spacing between the strips in the bottom layer is $\frac{L}{N+1}$.

- (c) Suppose a touch occurs at coordinates (i, j) for an arbitrary $N \times N$ touchscreen, and the voltage source and meter are connected as in the figures. A 5×5 example is shown in Figure 1(b). Find an expression for V_m as a function of V_s , N , i , and j . Again, the upper left corner is the coordinate $(1, 1)$ and the upper right coordinate is $(N, 1)$

- (d) Optional / Fun: Experiment with the Tinkercad models below to validate the theoretical results you just derived.

Tinkercad model of 2×2 equivalent circuit: <https://www.tinkercad.com/things/0wIXz3MkD7B>

Tinkercad model of 3×2 equivalent circuit: <https://www.tinkercad.com/things/k5oolj2tUEN>

2. Cell Phone Battery

As great as smartphones are, one of their drawbacks is that their batteries don't last a long time. For example, a Google Pixel phone, under typical usage conditions (internet, a few cat videos, etc.) uses 0.3W. We will model the battery as an ideal voltage source (which maintains a constant voltage across its terminals regardless of current) except that we assume that the voltage drops abruptly to zero when the battery is discharged (in reality the voltage drops gradually, but let's keep things simple).

Battery capacity is specified in mAh, which indicates how many mA of current the battery can supply for one hour before it needs to be recharged. The Pixel's battery has a battery capacity of 2770mAh at 3.8V. For example, this battery could provide 1000mA (or 3.8W) for 2.77 hours before the voltage abruptly drops from 3.8V to zero.

- (a) How long will a Pixel's full battery last under typical usage conditions?
- (b) How many coulombs of charge does the battery contain? How many usable electrons worth of charge are contained in the battery when it is fully charged? (An electron has 1.602×10^{-19} C of charge.)

- (c) Suppose the cell phone battery is completely discharged and you want to recharge it completely. How much energy (in J) is this? Recall that a J is equivalent to a W s.
- (d) Suppose PG&E charges \$0.12 per kWh. Every day, you completely discharge the battery (meaning more than typical usage) and you recharge it every night. How much will recharging cost you for the month of October (31 days)?
- (e) The battery has internal circuitry that prevents it from getting overcharged (and possibly exploding!). We will model the battery and its internal circuitry as a resistor R_{bat} . We now wish to charge the battery by plugging into a wall plug. The wall plug can be modeled as a 5 V voltage source and 200 m Ω resistor, as pictured in Figure 3. What is the power dissipated across R_{bat} for $R_{\text{bat}} = 1 \text{ m}\Omega$, 1 Ω , and 10 k Ω ? (i.e. how much power is being supplied to the phone battery as it is charging?). How long will the battery take to charge for each of those values of R_{bat} ?

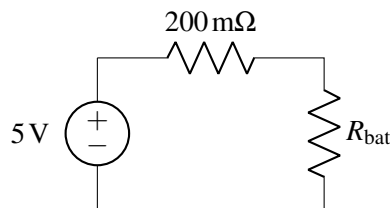


Figure 3: Model of wall plug, wire, and battery.